INCH-POUND

MIL-DTL-32546 /1(AS)
14 March 2016

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, RECEPTACLE, HIGH-SPEED DATA, THREADED, SQUARE FLANGE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32546.

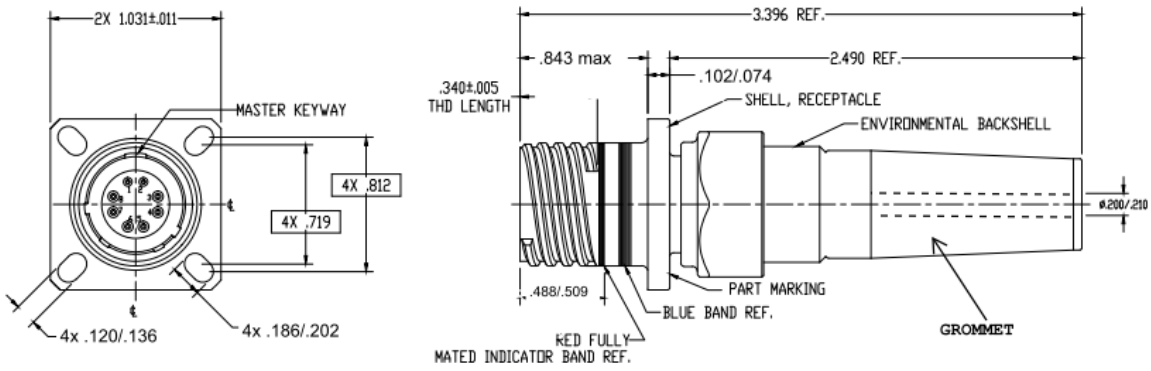


FIGURE 1. Receptacle, square flange, size 11.

TABLE I. Insert arrangement for size 11 shell.

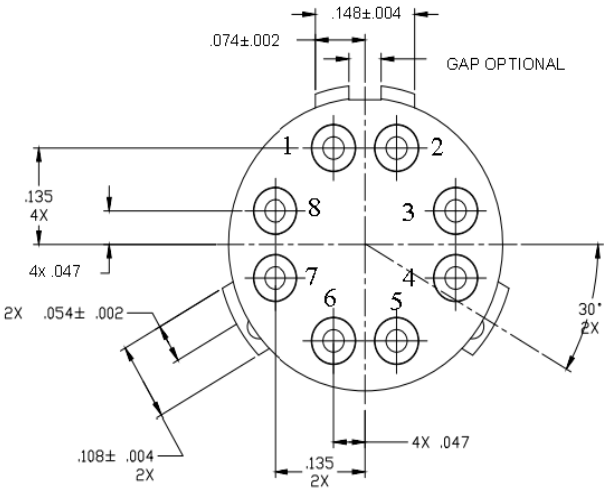
Insert Arrangement (see figure 2)	Service Rating (DWV)*
11-08	1300V rms**

*dielectric withstanding voltage
**root mean square

AMSC N/A

FSC 5935

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Front face

TABLE II. Contact location identification/Cable wire color		
Pin 1/	Cable Wire Colors 2/	EIA/TIA 568A/568B Pairs 3/
1	White/Blue	1
2	Blue	
3	White/Orange	2
4	Orange	
5	White/Green	3
6	Green	
7	White/Brown	4
8	Brown	

1/ Contact location identification is for reference only (see 3.40).
2/ Recommended cable wire color locations.
3/ Recommended EIA/TIA 586 pair allocation.

FIGURE 2. 11-08 insert detail.

TABLE III. Contact information.

Pin					
Type	Military Part No.	Crimp Tool 1/	Positioner 1/	Installing Tool 1/	Removal Tool 1/
22D Pin	M39029/58-360	M22520/2-01 M22520/7-01	M22520/2-09 M22520/7-07	M81969/14-01 M81968/8-01	M81969/14-01 M81969/8-02
	M39029/107-620				
	M39029/71-390, -391, -392	N/A		M81969/1-04 M81969/8-01	M81969/1-04 M81969/8-02

1/ See SAE AS39029/58, /71 and /107 for additional crimp tool and installation/removal tool information.

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TABLE III. Contact information - Continued.

Socket					
Type	Military Part No.	Crimp Tool 1/	Positioner 1/	Installing Tool 1/	Removal Tool 1/
22D Socket	M39029/56-348	M22520/2-01 M22520/7-01	M22520/2-09 M22520/7-07	M81969/14-01 M81968/8-01	M81969/14-01 M81969/8-02
	M39029/106-614				
	M39029/72-524, -525, -526	N/A		M81969/1-04 M81969/8-01	M81969/1-04 M81969/8-02

1/ See SAE AS39029/56, /72 and /106 for additional crimp tool and installation/removal tool information.

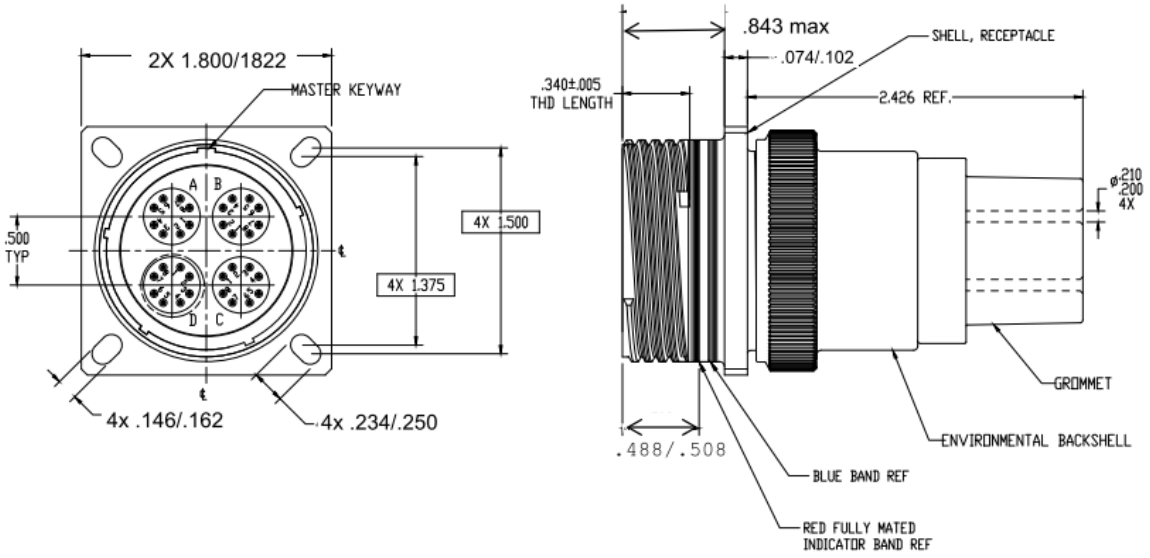


FIGURE 3. Receptacle, square flange, size 25.

TABLE IV. Insert layouts for size 25 shell.

Insert Arrangement (see figures 3 and 4)	Service Rating (DWV)
4 x 11-08	1300V rms

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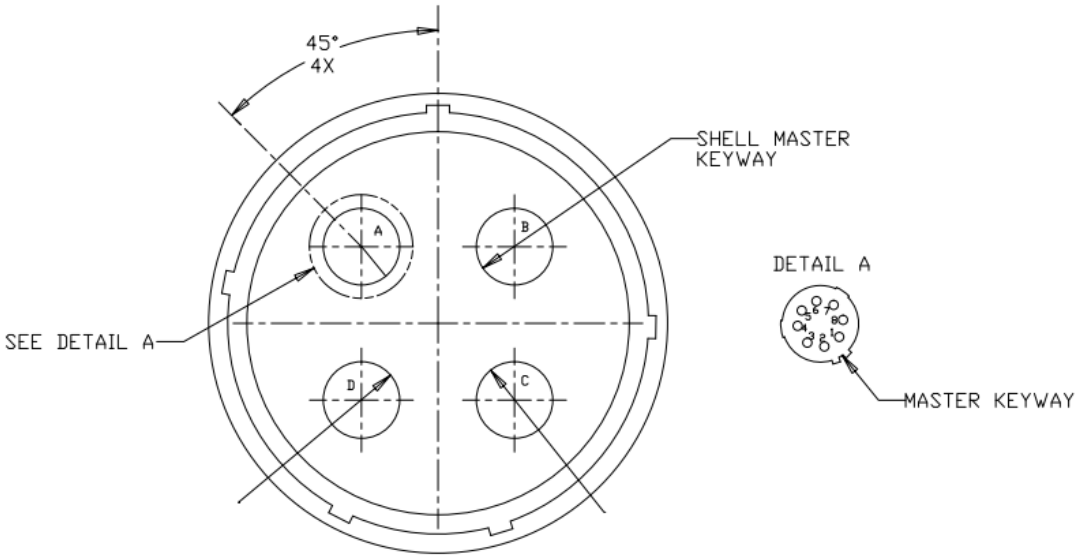


FIGURE 4. Insert orientation for size 25 shell. 1/

1/ Note that master key of insert shall be oriented at a 135 degree angle toward the center of the connector. Insert arrangement 11-08 shown in cavity A as reference.

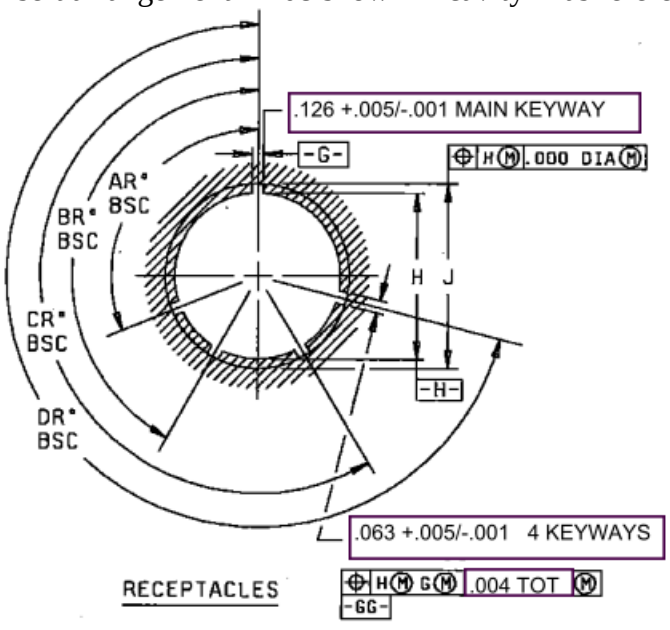


FIGURE 5. Receptacle polarization.

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TABLE V. Main key and keyway polarization.

Shell size	Key and keyway polarization identification letter	AR° or AP° BSC	BR° or BP° BSC	CR° or CP° BSC	DR° or DP° BSC
11	N	95	141	208	236
	A	113	156	182	292
	B	90	145	195	252
	C	53	156	220	255
	D	119	146	176	298
	E	51	141	184	242
25	N	80	142	196	293
	A	135	170	200	310
	B	49	169	200	244
	C	66	140	200	257
	D	62	145	180	280
	E	79	153	197	272

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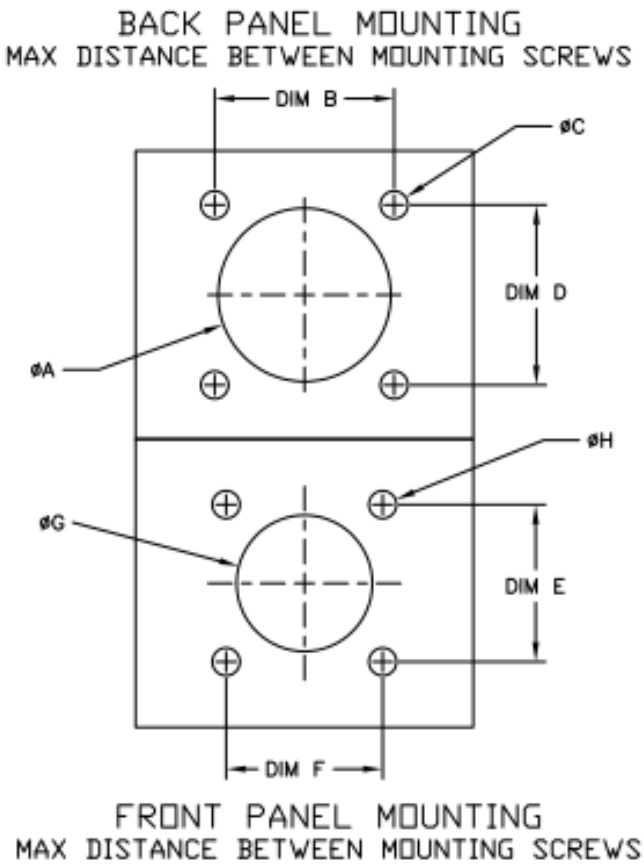


FIGURE 6. Panel mounting cutouts for square flanged receptacle.

TABLE VI. Panel mounting dimensions.

Shell size	A min.	B +/- .005	C +/- .005	D +/- .005	E +/- .005	F +/- .005	G min.	H +/- .005
11	.795	.812	.128	.812	.719	.719	.625	.128
25	1.672	1.500	.154	1.500	1.375	1.375	1.485	.154

NOTES:

1. Dimensions are in inches.
2. Environmental backshell included with connector kit.
3. Mating face of pin insert shown on insert layouts.

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REQUIREMENTS:

1. Materials:

Resilient inserts/backshell grommet – silicone elastomer per ASTM D6411/D6411M.
Plastic inserts – thermoplastic per ASTM D5205.

2. Dimensions and configuration: See Figures 1 through 6 and Tables I, and IV through VI. Interface dimensions shall be in accordance with connector intermateability. dimensions (series III) of MIL-DTL-38999.

3. This connector mates with MIL-DTL-32546/2.

4. Part or Identifying Number (PIN) example:

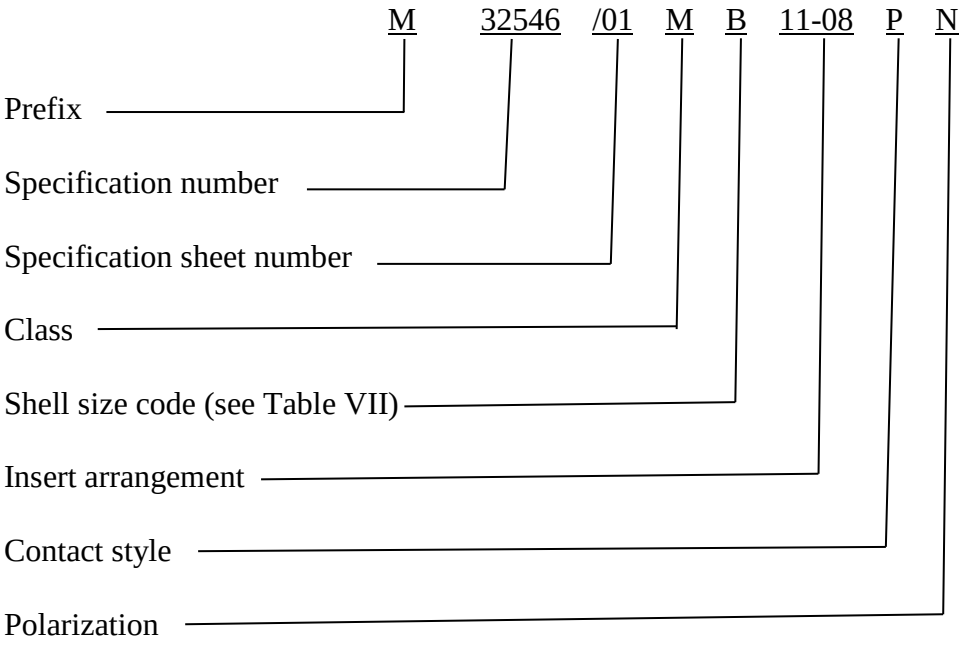


TABLE VII. Shell size code.

Shell size	Shell size code
11	B
25	F

5. See Table II for contact information for Figure 1 and 3 plug contacts. Contacts shall be available in pin or socket configurations.

6. The frequency range is 1-500 MHz.

7. Impedance is determined by the electrical performance and characteristics stated in MIL-DTL-32546.

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8. Connector physical requirements shall meet the requirements of MIL-DTL-38999, Series III, unless otherwise noted.

EXAMINATION AND TESTS:

1. Additional testing shall be performed as specified in Table VIII.

TABLE VIII. Cable braid retention test.

Inspection	Requirement	Test Paragraph
Cable braid retention	12.5 pounds force with slippage not to exceed 0.125 in.	SAE AS85049 cable pullout test (Method A ECIA EIA-364-38) except that the pull force is to be applied to the cable braid (less jacket).1/

1/ In the event of multiple cables, test applies to individual cables.

NOTES:

1. Intended use. The connectors qualified to this specification are intended for use as high speed data connectors using copper conductor for transmission of signals up to 10 gigabits per second (Gbps) such as IEEE 802.3 10GBASE-T, over four-pair cabling.

Referenced documents. In addition to MIL-DTL-32546 and its applicable specification sheets, this specification sheet references:

ASTM D6411/D6411M
ASTM D5205
ECIA EIA-364-38
EIA/TIA 586
IEEE 802.3
MIL-DTL-38999
SAE AS39029/56
SAE AS39029/58
SAE AS39029/71
SAE AS39029/72
SAE AS39029/106
SAE AS39029/107
SAE AS85049

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS
Project 5935-2015-156

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